

Most estimates

Unit one · more number of digits and estimation · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5A Unit 1 + Modern Education 5A Unit 2

Core traps: ● High frequency

The sub-test must be taken every year, and the estimation questions account for about 8-12% of Paper 1

Prerequisite knowledge: Class 1 (multi-digit reading and writing, decimal place value, rounding to approximate values)

Objective of this class: ❶ Understand the difference between estimation and actuarial science ❷ Master the segmentation method estimation ❸ K/M notation ❹ The strategy of estimating first and then calculating ❺ Estimation word problems

Our goals: ❶ Understand the difference between estimation and actuarial calculation ❷ Master the segmentation method for estimation ❸ K/M notation ❹ The strategy of estimating first and then calculating ❺ Estimation application questions

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question , 5 minutes)

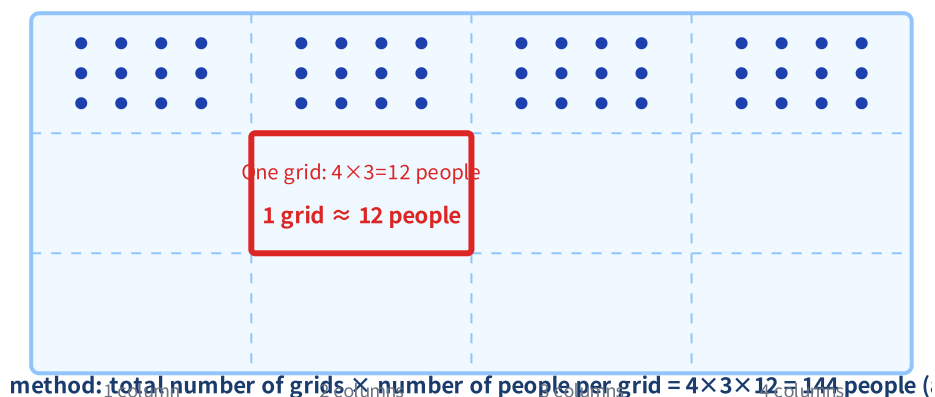
#	Question	Difficulty	Working Space (Show full working)
1	Round 38,562 to the nearest thousand.	Basic	
2	Write 4,725,000 as an approximate value in units of thousands (round to the nearest ten thousand).	Basic	
3	How much does 1K represent? How much does 1M represent? Write the numbers.	Basic	
4	Using rounding techniques, estimate the result of $398 + 612$ (rounded to the nearest hundred).	Advanced	
5	A photo shows the crowd: about 30 people horizontally and about 25 rows vertically. Use the segmentation method to estimate how many people are at the scene?	Advanced	

II.Core Knowledge + Worked Examples

Knowledge point 1: What is an estimate? Estimate $\neq$ Actuarial ● SSPA

- ① **Actuarial** |||SEP|||: Calculate the exact answer (for example: $398 + 612 = 1010$) Estimation |||SEP|||: Use approximate values to quickly find the "approximate" answer (for example: $400 + 600 \approx 1000$)
- ② ③ **Purpose of estimation: Quickly judge whether the answer is reasonable, used when precise numbers are not needed in daily life** Segmentation method |||SEP|||: Divide the large area into small grids $\rightarrow$ Count the number of people/objects in one grid $\rightarrow \times$ Number of grids = Estimated total number
- ⑤ Write " $\approx$ " (approximately equal) in front of the estimated answer, do not use "="
- ④ **segmentation method**: Divide a large area into small grids $\rightarrow$ count the number of people/objects in one grid $\rightarrow \times$ number of grids = estimated total number
- ⑤ Please write " $\approx$ " (approximately equal to) in front of the estimated answer, do not use "="

SVG: Schematic diagram of crowd estimation using segmentation method



Trap detonation example (the most important demonstration in this class)

The concert venue has approximately 28 seats per row, for a total of 32 rows. Use estimation methods to find approximately how many seats there are?

✗ Common mistakes (60% students)

$$28 \times 32 = 896$$

Direct actuarial calculation, no "estimation"!

✓ Correct solution (estimate first)

$$30 \times 30 \approx 900 \text{ pieces}$$

$28 \approx 30, 32 \approx 30 \rightarrow 30 \times 30 \approx 900$ (estimated answer)

 **Tip: "Use approximation for estimation, and equal sign for actuarial calculation; first take the approximate value, and then calculate the approximate number."**

 **The most frequent error: the question requires "estimation" but students did "actuarial calculation". When you see "approximately", "estimate" and "approximately" → be sure to use $\approx$!**

 **The second most frequent error: writing the "=" equal sign after estimating. The estimated result is not an exact value and must be written as $\approx$ (approximately equal to)!**

Knowledge Point - Worked Examples

#	Question	Difficulty	Working Space
Example 1	The picture below is divided into 24 cells, with about 15 people in each cell. Use the segmentation method to estimate the total number of people.		
Example 2	There are about 35 books on each shelf, for a total of 18 shelves. Use the estimation method to find approximately how many books there are? (rounded to the nearest ten)		

Knowledge point 2: K/M notation (concise representation of large numbers) SSPA required test

K and M are the internationally accepted "abbreviations for large numbers":

- ① **K = Thousand (Kilo):** 1K = 1,000 (for example: 5K = 5,000, 3.5K = 3,500)
- ② **M = Million (Mega):** 1M = 1,000,000 (for example: 2M = 2,000,000, 1.5M = 1,500,000)
- ③ **Common applications:** YouTube views (250K views), property prices (8.5M), population (7.5M)
- ④ **K → Number:** Multiply 1,000 (decimal point shifted 3 places to the right)
- ⑤ **M → Number:** Multiply by 1,000,000 (decimal point moved 6 places to the right)

K

Kilo

1K = 1,000
 $\times 1,000$

M

Mega

1M = 1,000,000
 $\times 1,000,000$

→

K to digital

$\times 1,000$ Decimal point shifted
3 places to the right

→

M to digital

$\times 1,000,000$ Decimal point
shifted 6 places to the right

example

Example 3: Write 45K and 2.5M as ordinary numbers.

example

Example 4: A video has 1,250,000 views, expressed in two ways: K and M.

Knowledge point 3: The strategy of estimating first and then calculating (sequence trap!) ● SSPA Advanced

Core Principle: Approximate first, do calculations later – don't do calculations first and then do approximations!

- ① **Correct order** |||SEP|||: original data → take approximation → calculation → estimated result
Wrong order |||SEP|||: original data → actuarial → take approximation (the meaning of estimation is lost, and the answer may be different!)
- ② **Why "estimate first"?** Because rounding before calculation is different from rounding after actuarial calculation, the result
- ③ **may be different** ④ Sub-test trap: The question says "Estimate first, then calculate" ↔ Two steps must be shown, no skipping **may be different !**
- ④ Sub-test trap: The question says "Estimate first, then calculate" ↔ Two steps must be shown, no skipping.

Trap detonation example

The fast-food restaurant sold 387 burgers and 614 sodas throughout the day. Estimate first and then calculate how many servings of food were sold in total?

✗ Calculate first and then approximate

$$387+614=1001 \approx 1000$$

Wrong order! The question requires "Estimate first and then calculate"

✓ Estimate first and then calculate

$$\text{Estimate: } 400+600 \approx 1000$$

$$\text{Actuarial: } 387+614=1001$$

First make an approximate estimate and then do an actuarial check

⚠ **SSPA killer question: "Estimate first, calculate later" - You must write $\approx$ that step first, and the order must not be reversed!**

⚠ **Unit trap: When estimating, make sure the units are consistent (convert all to the same units before estimating), do not mix K, M, and ordinary numbers!**

Knowledge Point 2 + 3 Worked Examples

#	Question	Difficulty	Working Space
Example 5	There were 23,500 people attending the concert, denoted by K (rounded to one decimal place).		
Example 6	Three items from the supermarket: \$28.5, \$43.2, \$19.8. Estimate the total first (round to the nearest ten dollars), then calculate the exact total.		
Example 7	The minibuses can carry up to 16 people and the school has 318 pupils on the trip. What is the estimated minimum number of minibuses needed? (Hint: $318 \approx 320$, $320 \div 16 = ?$)		

**Example
8**

Population of three cities: A=2.8M, B=1,250K, C=3,100,000. (a) Convert all to M and express (b) Estimate the approximate total population of the three cities, M (rounded to one decimal place).



III. Lesson Layered Synchronization Practice

Knowledge point 4: Estimation application questions (subject to sub-test text questions) SSPA compulsory exam

Four-step method for solving application problems on estimation:

- ① **Circle keywords**("approximately" "estimate" "approximately" → use $\approx$; "total" "total" → addition/multiplication)
- ② **Take approximate values**(according to the accuracy required by the question, such as rounding to the nearest hundredth/thousandth place)
- ③ **Estimate calculation**(Use approximate values to calculate, write $\approx$ before the answer)
- ④ **If necessary, do actuarial comparison**(Some questions require "estimate first and then calculate")

Basic layer (total 7 questions, everyone must do)

#	Question	Difficulty	Working Space
1	Write 68K as an ordinary number.		
2	Write 3.2M as an ordinary number.		
3	Estimate using rounding method $287 + 514$ (rounded to the nearest hundred).		
4	A picture is divided into 18 boxes, with approximately 22 people in each box. Use the segmentation method to estimate the total number of people. (rounded to the nearest hundred)		
5	Let K and M represent 4,800,000 respectively.		
6	Write the result of $1.2K + 800$. (Tip: Convert 1.2K to an ordinary number first and then calculate)		
7	Estimate using rounding method $495 \div 7$ (rounded to the nearest ten).		

Advanced layer (total 5 questions,   choose do)

#	Question	Difficulty	Working Space
8	One book has 238 pages and the other has 476 pages. First estimate the approximate number of pages in the two books (rounded to the		

	nearest hundred), and then calculate the exact total number of pages.		
9	The concert sold 1.25K tickets and the other 0.85K tickets. How many copies were sold in the two games? (Use K to estimate first, then convert to numbers)		
10	A certain housing estate in Hong Kong has eight buildings, each with approximately 320 units. Use an estimate to find out how many units there are in the housing estate? (rounded to the nearest thousand)		
11	There are 1,250 candies in a box, and there are 6 boxes available. First approximate the total sugar count (rounded to the nearest thousand) and then calculate the exact total.		
12	The school has 4 levels with 148 students in each level. First estimate the approximate number of people in the school (rounded to the nearest hundred), and then make an arithmetic calculation. What is the difference between estimation and actuarial calculation? Why?		

 **challenge layer (total 3 questions,  choose do, SSPAKiller Questions)**

#	Question	Difficulty	Working Space
13	The views of the three videos are 2.4M, 1.85M and 0.95M respectively. First estimate the total number of views (rounded to M to one decimal place), then calculate the exact total.		
14	Xiao Ming said: " $398+612 \approx 400+600 \approx 1000$, so the answer is about 1000." Xiao Hua said: "No! $398+612=1010 \approx 1000$, the answer is the same!" Whose statement meets the definition of "estimate"? Why?		
15	The school auditorium can accommodate 850 people, and there are currently 23 classes with about 32 people in each class. Use estimates to determine whether the auditorium is large enough to accommodate the entire school population? If not, how many seats are still owed? (Tip: Estimate the total number of people first, then compare to 850)		

estimationapplicationquestion (all must do, must display "estimationstep")

#	Question	Difficulty	Working Space
16	Stationery store pencils cost \$3.80 each and ball pens cost \$5.20 each. Xiao Ming bought 10 pencils and 10 ball pens each. Estimate the total first (round to the nearest ten dollars), then calculate the exact total.		

17	The school auditorium has 28 seats per row and 36 rows in total. Use the estimation method to find approximately how many seats there are in the auditorium? (rounded to the nearest hundred)		
18	Attendance at Hong Kong Disneyland on a given day is about 34,500. Let K represent the number of attendees. (Rounded to one decimal place)		
19	A housing estate has 12 blocks, each with 38 floors and 6 units per floor. Use the estimation method to find out approximately how many units there are in a housing estate? (Tip: estimate in two steps)		
20	The mall's three-day attendance: 8,750 on Friday, 15,200 on Saturday and 13,800 on Sunday. First estimate the total number of visitors for the three days (rounded to the nearest thousand), and then calculate the exact total.		
21	A certain YouTuber's three most popular videos have been viewed: 1.2M, 850K, and 2.35M. Question: (a) How many M are the total views of the three videos? (b) What is the precise total number of views?		

IV. Ultimate challenge area

#	Question	Difficulty	Working Space
 1	A large concert venue is divided into three areas: A, B, and C. Zone A has 4,850 people, Zone B has 3,920 people and Zone C has 6,180 people. (a) First estimate the total number of people in the venue (rounded to the nearest thousand). (b) Calculate the total number of people in the venue. (c) How much does the estimated result differ from the actuarial result? Is this gap reasonable? Why?		
 2	City A has a population of approximately 3.5M, City B has a population of approximately 2,800K, and City C has a population of approximately 1,250,000. (a) Let M represent the population of all three cities. (b) What is the total population of the three cities, M? (c) If the three cities are merged, is it more appropriate to use M or K to express the total population? Why?		
 3	Xiaowen bought 3 items in the supermarket: \$12.8, \$23.5, and \$9.9. (a) First estimate the total (round to the nearest ten dollars). (b) Calculate the exact total. (c) Compare estimates and actuarial differences. (d) If each item is approximated first and then approximated, will the results be different from those first added and then approximated? Use these three numbers as an example.		

V. Class afterhomework

Basic must-do questions (total 6 questions, must write estimation step)

#	Question	Difficulty	Working Space
H1	Write 125K as an ordinary number.		
H2	Write 5.5M as an ordinary number.		
H3	Estimate using rounding method $613 + 289$ (rounded to the nearest hundred).		
H4	The parking lot is divided into 24 bays, with approximately 18 cars in each bay. Use the segmentation method to estimate the total number of vehicles. (rounded to the nearest hundred)		
H5	Let's first express 3,750,000 by K and then by M.		
H6	The games had 2,380 people in the morning and 1,720 in the afternoon. First estimate the total number of people for the day (rounded to the nearest thousand), and then calculate the exact total.		

For advanced, choose doquestion (total 2 questions,  choose do)

#	Question	Difficulty	Working Space
H7	The satellite images of the three cities each have 48 grids, 52 grids, and 36 grids, and each grid has about 25 buildings. Use estimates to find out how many buildings there are in the three cities?		
H8	The number of downloads of a certain app in the world: 12.5M in Asia, 8.3M in Europe, 15.7M in the Americas, and 3.9M in other regions. Question: (a) How many M are the total global downloads? (b) How many more M are there in Asia than in Europe?		

VI. The Lessoncorecommon errorssummary

☒ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

 Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	Estimate $\neq$ Actuarial SEP : The question requires estimation but actuarial calculation is done, or the estimated answer uses the "=" equal sign: The question requires an estimate but an actuarial calculation is done, or the answer to the estimate uses the "=" equal sign.	Use " $\approx$ " for estimation and "=" for actuarial calculation.
2	K/M unit confusion SEP : 1K = 1,000 but 1M = 1,000,000 (cannot be used as 10,000!): 1K = 1,000 but 1M = 1,000,000 (cannot be used as 10,000!)	1K = $\times 1,000$, 1M = $\times 1,000,000$
3	Actuarial first and then get approximate values SEP : The order is reversed and the requirement of "estimate first, then calculate" is forgotten. The order is reversed and the requirement of "estimate first, then calculate" is forgotten.	First take the approximate value $\rightarrow$ calculate $\rightarrow$ get the estimated result
4	Decimal point shift error during K/M conversion SEP : For example, 2.5K = 250 (should be 2,500): e.g. 2.5K = 250 (should be 2,500)	Decimal point shifted to the right: K=3 digits, M=6 digits
5	The number of people per square in the segmentation method is not rounded SEP : Use precise numbers to count the number of people in each square (for example, 12.3 people per square): Use precise numbers to count the number of people per square (for example, 12.3 people per square)	The number of people in each square should also be an approximate integer
6	Mixing different units SEP : K + ordinary numbers are added directly (for example, 3K + 500): K + ordinary numbers are added directly (for example, 3K + 500)	Unify the units first (convert all to numbers or all to K/M)